The same covering bound holds for maximal families of perfect matchings

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Editorial note. The counted statement is the author's labelled environment at source0.tex lines 742-750 and its proof environment opens at line 751; the pairing is by adjacency and the retained proof range is the whole environment.

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Editorial scope. One original result of the pinned author manuscript, counted once: the coset covering bound for degree-two maximal families of perfect matchings (author label lem:main-ub-pms).

Allowed standard prerequisites (editorial).

- Maximal 2-intersecting family. The family is maximal, so it cannot be enlarged while keeping the intersection property.. It replaces the degree bound of the permutation case.
- Certificate size C_1(f). The largest size of a common certificate over the members, for the matching setting.. It measures the cosets used in the covering.
- Matching counterparts of the three tools. The representative, covering and structure lemmas all have perfect-matching versions with the same shape.. They carry the iteration over to matchings.

Lemma 0.1. *Let $f\colon \mathcal{M}_{2n} \to \{0,1\}$ be the characteristic function of a 2-intersecting family, and suppose that $C(f) \leq n-2$. If $f(m_1) = f(m_2) = 1$ and C_{m_1}, C_{m_2} are minimum certificates for m_1, m_2 (respectively), then $|C_{m_1} \cap C_{m_2}| \geq 2$.*

Lemma 0.2. *Let $\mathcal{F} \subseteq S_n$ be a 2-intersecting family whose characteristic function f has degree at most 2. Suppose that $C(f) \leq n-2$.*

The family $\mathcal{F}$ is contained in the union of at most T many $C(f)$-cosets, where

$$T = \frac{2\lfloor C(f)/2 \rfloor (C(f)-1)!}{2^{\lfloor C(f)/2 \rfloor}}.$$

Moreover, $C(f) \geq 2$.

Lemma 0.3. *Let f be the characteristic function of a 2-intersecting family $\mathcal{F} \subseteq \mathcal{M}_{2n}$, and suppose that $C(f) \leq n-2$.*

If $m \in \mathcal{F}$ then $\mathcal{F}$ is r-covered for $r = C(f,m) - 1$.

Lemma 0.4. *Let $\mathcal{F} \subseteq \mathcal{M}_{2n}$ be an r-covered family, say $\mathcal{F} \subseteq [i_1 \leftrightarrow j_1] \cup \cdots \cup [i_r \leftrightarrow j_r]$, where $r \leq n-3$. Suppose that $\mathcal{F}$ is maximal, and that its characteristic function f has degree at most 2.*

Every $m \in \mathcal{F}$ has a certificate of the form $\big\{\{i_1, m(i_1)\}, \ldots, \{i_r, m(i_r)\}, \{i,j\}\big\}$.

Lemma 0.5. *Let $\mathcal{F} \subseteq \mathcal{M}_{2n}$ be a maximal 2-intersecting family whose characteristic function f has degree at most 2, and suppose that $C(f) \leq n-2$. Let $C_1(f)$ the maximum certificate complexity of any $m \in \mathcal{F}$.*

The family $\mathcal{F}$ is contained in the union of at most T many $C_1(f)$-cosets, where

$$T = \frac{2\lfloor C_1(f)/2 \rfloor (C_1(f)-1)!}{2^{\lfloor C_1(f)/2 \rfloor}}.$$

Moreover, $C_1(f) \geq 2$.

Proof. Let $\mu \in \mathcal{F}$ be a perfect matching satisfying $C(f,\mu) = C_1(f)$. Theorem 0.1, applied to $m_1 = m_2 = \mu$, shows that $C(f,\mu) \geq 2$, and so $C_1(f) \geq 2$. Theorem 0.3 shows that $\mathcal{F}$ is r-covered, where $r = C(f,\mu) - 1$. Note that $r \leq C(f) - 1 \leq n-3$. Without loss of generality, we can assume that $\mathcal{F} \subseteq [1 \leftrightarrow r+1] \cup [2 \leftrightarrow r+2] \cup \cdots \cup [r \leftrightarrow 2r]$.

Theorem 0.4 shows that every $m \in \mathcal{F}$ has a certificate of the form $\big\{\{1, m(1)\}, \ldots, \{r, m(r)\}, \{i,j\}\big\}$, which we call the *standard certificate*. The theorem also implies that $\mathcal{F}$ is not $(r-1)$-covered: otherwise, every perfect matching in $\mathcal{F}$ would have a certificate of size $r < C_1(f)$, which contradicts the definition of $C_1(f)$.

A standard certificate $\big\{\{1, m(1)\}, \ldots, \{r, m(r)\}, \{i,j\}\big\}$ satisfies $m(1), \ldots, m(r) \notin [r]$, since otherwise Theorem 0.4 would imply that $\mathcal{F}$ is $(r-1)$-covered (since some of the pairs $\{t, m(t)\}$ coincide), which is impossible. This allows us to color the pairs in a standard certificate unambiguously: a pair $\{t, m(t)\}$ is colored t, and a pair $\{i,j\}$ with $i, j \notin [r]$ is colored $r+1$. Every standard certificate contains exactly one pair of each color.

The remainder of the proof is identical to that of Theorem 0.2. □

References

[AK97] Rudolf Ahlswede and Levon H. Khachatrian. The complete intersection theorem for systems of finite sets. *European J. Combin.*, 18(2):125–136, 1997.

[AK99] Rudolf Ahlswede and Levon H. Khachatrian. A pushing-pulling method: new proofs of intersection theorems. *Combinatorica*, 19(1):1–15, 1999.

[BdW02] Harry Buhrman and Ronald de Wolf. Complexity measures and decision tree complexity: a survey. *Theoret. Comput. Sci.*, 288(1):21–43, 2002. Complexity and logic (Vienna, 1998).

[BMRR21] Angelot Behajaina, Roghayeh Maleki, Aina Toky Rasoamanana, and A. Sarobidy Razafimahatratra. 3-setwise intersecting families of the symmetric group. *Discrete Math.*, 344(8):Paper No. 112467, 15, 2021.

[CK03] Peter J. Cameron and C. Y. Ku. Intersecting families of permutations. *European J. Combin.*, 24(7):881–890, 2003.

[CSST08] Tullio Ceccherini-Silberstein, Fabio Scarabotti, and Filippo Tolli. *Harmonic analysis on finite groups*, volume 108 of *Cambridge Studies in Advanced Mathematics*. Cambridge University Press, Cambridge, 2008. Representation theory, Gelfand pairs and Markov chains.

[Daf22] Neta Dafni. Complexity measures on the symmetric group and beyond. Master's thesis, Technion — Israel Institute of Technology, 2022.

[DFL+21] Neta Dafni, Yuval Filmus, Noam Lifshitz, Nathan Lindzey, and Marc Vinyals. Complexity measures on the symmetric group and beyond. In *12th Innovations in Theoretical Computer Science Conference*, volume 185 of *LIPIcs. Leibniz Int. Proc. Inform.*, pages Art. No. 87, 5. Schloss Dagstuhl. Leibniz-Zent. Inform., Wadern, 2021.

[EFP11] David Ellis, Ehud Friedgut, and Haran Pilpel. Intersecting families of permutations. *J. Amer. Math. Soc.*, 24(3):649–682, 2011.

[EKR61] P. Erdős, Chao Ko, and R. Rado. Intersection theorems for systems of finite sets. *Quart. J. Math. Oxford Ser. (2)*, 12:313–320, 1961.

[Ell12a] David Ellis. A proof of the Cameron-Ku conjecture. *J. Lond. Math. Soc. (2)*, 85(1):165–190, 2012.

[Ell12b] David Ellis. Setwise intersecting families of permutations. *J. Combin. Theory Ser. A*, 119(4):825–849, 2012.

[FD77] Péter Frankl and Mikhail Deza. On the maximum number of permutations with given maximal or minimal distance. *J. Combinatorial Theory Ser. A*, 22(3):352–360, 1977.

[FHKL16] Yuval Filmus, Hamed Hatami, Nathan Keller, and Noam Lifshitz. On the sum of the L_1 influences of bounded functions. *Israel J. Math.*, 214(1):167–192, 2016.

[Fil17] Yuval Filmus. A comment on intersecting families of permutations. arXiv:1706.10146, 2017.

[FMS21] Shaun Fallat, Karen Meagher, and Mahsa N. Shirazi. The Erdős-Ko-Rado theorem for 2-intersecting families of perfect matchings. *Algebr. Comb.*, 4(4):575–598, 2021.

[GM09] Chris Godsil and Karen Meagher. A new proof of the Erdős-Ko-Rado theorem for intersecting families of permutations. *European J. Combin.*, 30(2):404–414, 2009.

[GM16] Chris Godsil and Karen Meagher. *Erdős-Ko-Rado theorems: algebraic approaches*, volume 149 of *Cambridge Studies in Advanced Mathematics*. Cambridge University Press, Cambridge, 2016.

[GM17] Chris Godsil and Karen Meagher. An algebraic proof of the Erdős-Ko-Rado theorem for intersecting families of perfect matchings. *Ars Math. Contemp.*, 12(2):205–217, 2017.

[Lin17] Nathan Lindzey. Erdős-Ko-Rado for perfect matchings. *European J. Combin.*, 65:130–142, 2017.

[Lin18a] Nathan Lindzey. Intersecting families of permutations of perfect matchints. arXiv:1811.06160, 2018.

[Lin18b] Nathan Lindzey. *Matchings and Representation Theory*. PhD thesis, University of Waterloo, 2018.

[LM04] Benoit Larose and Claudia Malvenuto. Stable sets of maximal size in Kneser-type graphs. *European J. Combin.*, 25(5):657–673, 2004.

[MM05] Karen Meagher and Lucia Moura. Erdős-Ko-Rado theorems for uniform set-partition systems. *Electron. J. Combin.*, 12:Research Paper 40, 12, 2005.

[MR21] Karen Meagher and A. Sarobidy Razafimahatratra. The Erdős-Ko-Rado theorem for 2-pointwise and 2-setwise intersecting permutations. *Electron. J. Combin.*, 28(4):Paper No. 4.10, 21, 2021.

[NÖ03] Sampo Niskanen and Patric R. J. Östergård. Cliquer user's guide, version 1.0. Technical Report T48, Communications Laboratory, Helsinki University of Technology, Espoo, Finland, 2003.

[O'D14] Ryan O'Donnell. *Analysis of Boolean functions*. Cambridge University Press, New York, 2014.

[Sag01] Bruce E. Sagan. *The symmetric group*, volume 203 of *Graduate Texts in Mathematics*. Springer-Verlag, New York, second edition, 2001. Representations, combinatorial algorithms, and symmetric functions.

[Sag22] Sage Developers. *SageMath, the Sage Mathematics Software System (Version 9.5)*, 2022. `https://www.sagemath.org`.